

Equity in Small Business Lending:

Predicting Loan Default and Auditing Fairness in SBA-Guaranteed Loans

Final Report

CPSC 5071 — Data Management for Data Science — Winter 2026

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GitHub: <https://github.com/CPSC-5071-SELECT-ALL-FROM-STRUGGLING/FairLend-SBA>

1. Introduction

Small businesses are fundamental to the U.S. economy, and the U.S. Small Business Administration (SBA) supports them by guaranteeing loans to reduce lender risk [1]. However, when a guaranteed loan defaults, the financial loss is shared between the lending bank and U.S. taxpayers. Between 1987 and 2014, approximately 17.2% of SBA-guaranteed loans in our dataset were charged off [2].

At the same time, research has documented persistent disparities in access to capital for minority-owned and women-owned businesses [3]. Federal Reserve survey data confirm that these gaps persist, with minority-owned firms more likely to be denied credit or discouraged from applying [4]. As financial institutions increasingly adopt data-driven decision-making tools, a critical question emerges: could predictive models used in lending inadvertently encode or amplify these existing disparities?

This project addresses two research questions:

Predictive: Can we predict SBA loan default based on loan characteristics, business attributes, and macroeconomic conditions?

Inferential: Do loan outcomes differ across industry-state segments with varying concentrations of women-owned and minority-owned businesses? If deployed, would a predictive model produce equitable error rates across demographic segments?

Our analysis demonstrates that loan default can be predicted with high accuracy (AUC = 0.974) using a HistGradientBoosting classifier, with loan term, approval year, and macroeconomic indicators as the dominant predictors. A Logistic Regression baseline (AUC = 0.873) confirms through inferential analysis that once economic pressures and firm scale are controlled for, the default disparity observed in minority-dense segments effectively disappears. The fairness audit reveals that demographic composition features contribute almost nothing to predictions (ranking 23rd and 30th out of 34 features by permutation importance), and that error rates are substantially equitable across demographic segments — with all women-ownership gaps under 0.2% and a 6.3% FNR gap for minority-ownership that actually favors high minority-ownership sectors. However, the inferential analysis also reveals a residual structural disparity: loans in high-minority-density segments carry approximately 4x higher default odds even after controlling for all business characteristics, suggesting that algorithmic fairness alone is insufficient without complementary policy interventions addressing the structural economic conditions underlying these disparities.

2. Data and Data Management

2.1 Data Sources

We integrated three primary data sources and one validation source to construct the modeling dataset:

Source	Description	Key Variables	Size	Join Key
SBA Loan Data (Kaggle)	SBA-guaranteed loans, 1987–2014	Loan amount, term, NAICS, state, default status	899,164 loans	NAICS + State + Year
Census ABS (API)	Industry × State demographic composition	pct_women_owned, pct_minority_owned, pct_vet by sector	1,024 segments	NAICS + State
FRED (API)	Monthly macroeconomic indicators, 1987–2014	Fed Funds Rate, Prime Rate, CPI, Unemployment, Bank Spread	28 annual rows	Year
Fed SBCS (Validation)	Firm-level survey on financing outcomes	Owner demographics, approval/denial rates	~7,600 firms	Reference only

The SBA loan dataset provides the core transaction-level records with 27 columns covering loan characteristics, borrower attributes, and repayment outcomes. Census ABS data provides industry-by-state demographic composition of business ownership, collected via the Census Bureau API for 2002, 2007, and 2012 survey years and averaged to define 1,024 industry-state segments. FRED data provides macroeconomic context through monthly time series aggregated to annual averages, including derived features such as bank spread and real interest rate. The Federal Reserve Small Business Credit Survey serves as an independent validation source for ecological-level patterns.

2.2 Data Profiling and Cleaning

We profiled the raw SBA dataset (899,164 rows × 27 columns) and identified 10 data quality issues requiring remediation:

#	Issue	Columns	Rows Affected	Action
1	Missing target variable	MIS_Status	1,997 (0.2%)	Dropped rows
2	Target has trailing spaces	MIS_Status	All rows	Stripped whitespace
3	Currency stored as strings	DisbursementGross, GrAppv, SBA_Appv, BalanceGross, ChgOffPrinGr	All rows	Removed \$ and commas, converted to float

4	Missing industry codes	NAICS	~200K (22.5%)	Hybrid: state-average fallback
5	Dirty categorical values	RevLineCr, LowDoc	~120K	Non-Y/N values set to NULL
6	Unknown NewExist	NewExist	~378	Value 0 set to NULL
7	Pre-1987 sparse data	ApprovalFY	~12K	Filtered to 1987+
8	Invalid year entry	ApprovalFY	1 row	Coerced to NaN, dropped
9	Split NAICS codes	NAICS 2-digit	~101K	Mapped to Census sector ranges via SQL views
10	Date columns as strings	ApprovalDate, DisbursementDate	All rows	Parsed for year extraction

After cleaning, the SBA dataset was reduced to 886,649 rows × 29 columns (two derived columns added: `is_default` and `naics_2digit`). The overall default rate of 17.2% was preserved, confirming that no systematic bias was introduced by the cleaning process.

For the Census ABS data, we addressed cross-tabulation double-counting by filtering to sector-level NAICS codes and pinning the sex dimension to “All firms” when computing minority-owned percentages, and the race dimension to “All firms” when computing women-owned percentages. The resulting demographic segments were averaged across 2002–2012 Census snapshots to produce stable estimates for each industry-state combination.

FRED data required aggregation from monthly to annual frequency, computing mean values for each indicator per year. Additional derived features were computed including bank spread (prime rate minus federal funds rate) and real interest rate (prime rate minus inflation).

2.3 Database Schema and Integration

We implemented a SQLite relational database with separate schemas optimized for each data source. The Census data uses a star schema with a central fact table surrounded by dimension entities, enabling efficient aggregation across industry sectors, states, and years. FRED data uses a time series schema with yearly aggregated indicators. The SBA loan data serves as the central transaction table.

SQL views were created for aggregation and pivoting: a segment-level view (`V_CENSUS_BUSINESS_SEGMENTS`) computes demographic percentages by industry-state, and a state-level view (`V_CENSUS_BUSINESS_SEGMENTS_STATE`) computes state-wide averages across all industries. A hybrid view (`V_CENSUS_BUSINESS_SEGMENTS_V2`) uses industry-state demographics when available and falls back to state averages for loans with unclassified industry codes, maximizing coverage from 54% to nearly 100% of all loans.

The final modeling dataset was constructed by merging SBA loans with Census demographic segments, FRED macroeconomic indicators, and engineered features including relative employment size, loan-to-segment revenue ratios, and minority density group classifications. This produced 886,649 rows with 36 columns and 34 modeling features. Demographic coverage

is approximately 100% under the hybrid approach, compared to 54% when using industry-state matching alone.

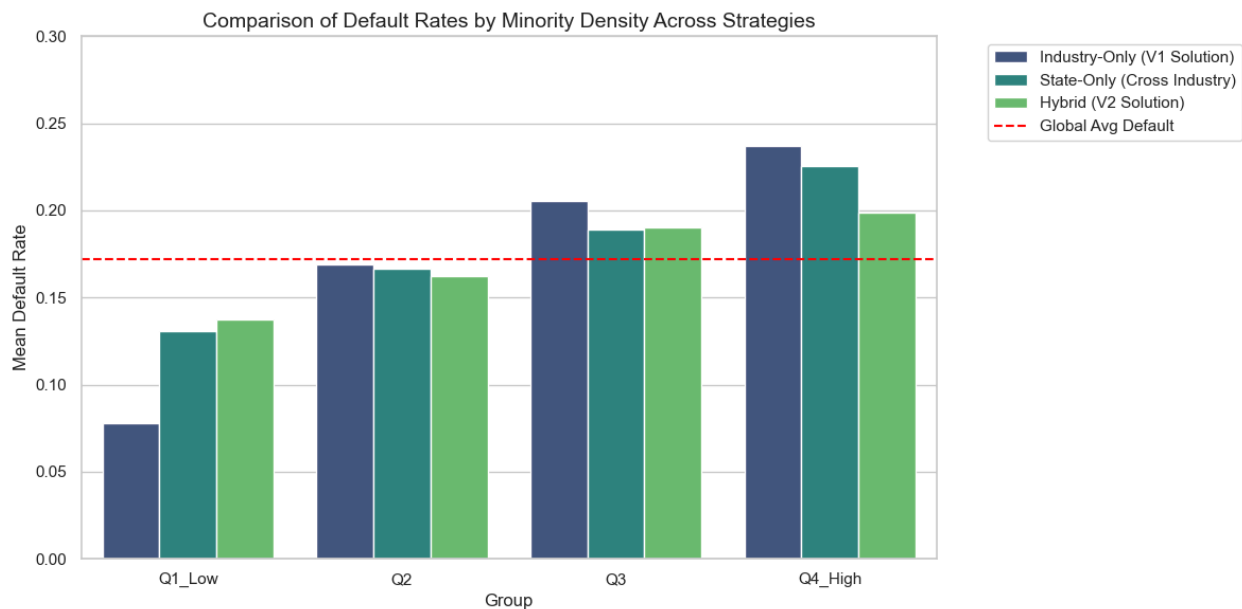


Figure 1: Comparison of default rates by minority density across demographic join strategies

3. Fairness Analysis

3.1 Model Performance Summary

We trained three models on the 886,649-loan dataset using an 80/20 stratified train/test split (random_state=42). The two ensemble models used a shared preprocessing pipeline with median imputation and standard scaling for numeric features, and constant-value imputation with one-hot encoding for categorical features. The Logistic Regression used ordinal encoding and was tuned with balanced class weights to prioritize default detection.

Metric	Random Forest	HistGradientBoosting	Logistic Regression
Test AUC-ROC	0.946	0.974	0.873
Precision (default)	0.858	0.869	0.410
Recall (default)	0.680	0.810	0.850
F1 (default)	0.759	0.839	0.550

HistGradientBoosting is the strongest model, achieving a 0.974 AUC with the best balance of precision and recall. Random Forest underperforms on recall (68.0%), missing nearly a third of defaults. Logistic Regression, while achieving the highest recall (85%), does so at the cost of very low precision (41%) — it flags many non-defaulting loans as defaults. However, the Logistic Regression serves a critical inferential role: its coefficients reveal that macroeconomic indicators (federal funds rate coefficient = 13.2, prime rate = 7.1) overwhelmingly dominate the prediction of default, far outweighing any demographic features.

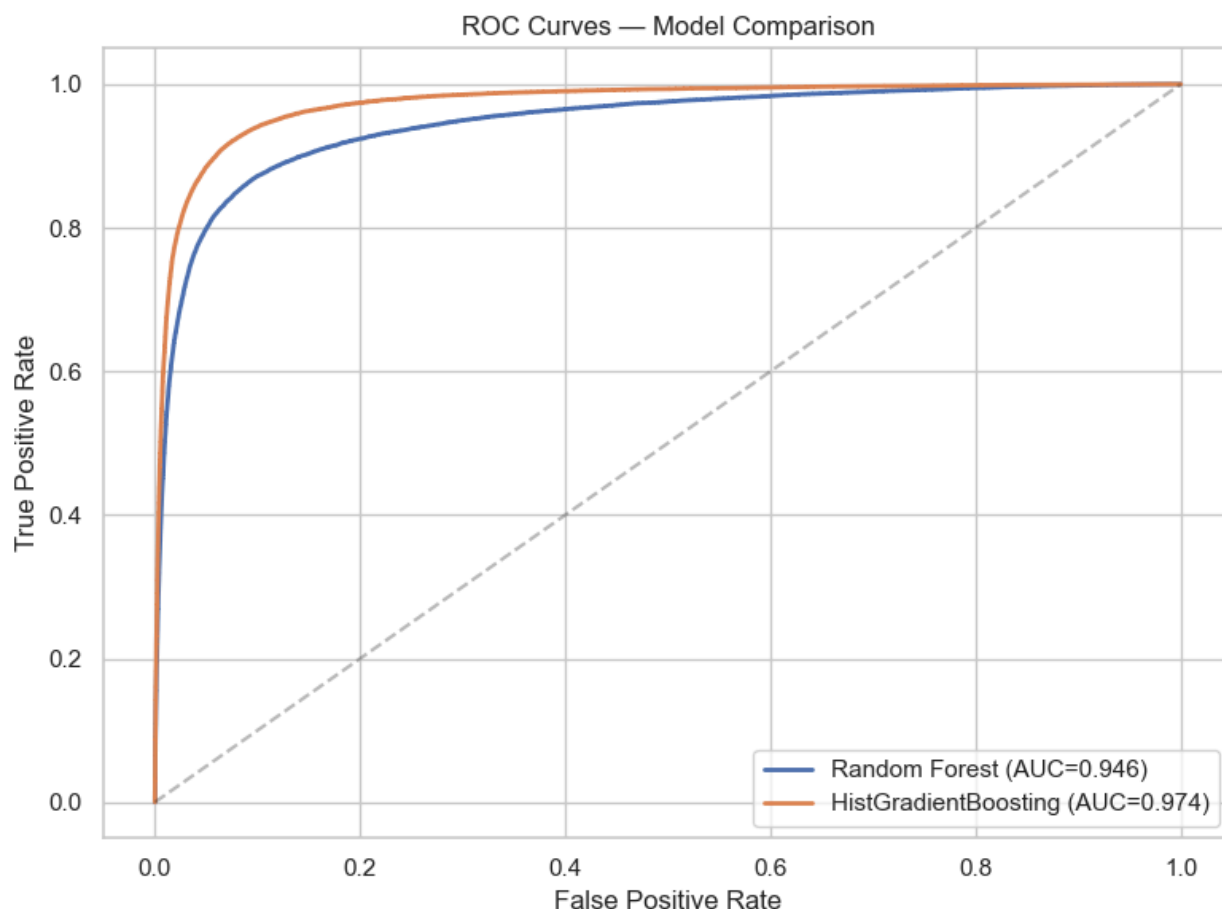


Figure 2: ROC curves — model comparison

3.2 Fairness Audit Methodology

To assess whether the models would produce equitable outcomes if deployed, we conducted a fairness audit following a median-split subgroup approach. We divided the test set into demographic subgroups based on the Census-derived ownership composition of each loan’s industry-state segment:

Women-ownership groups: Loans in industry-state segments with `pct_women_owned` at or above the median (1.693) were classified as “High Women-Ownership”; those below as “Low Women-Ownership.”

Minority-ownership groups: Loans in segments with `pct_minority_owned` at or above the median (3.985) were classified as “High Minority-Ownership”; those below as “Low Minority-Ownership.”

We computed False Positive Rate (FPR) and False Negative Rate (FNR) for each subgroup. FPR measures how often the model incorrectly flags a non-defaulting loan as a default (penalizing good borrowers), while FNR measures how often the model misses actual defaults (exposing lenders to undetected risk). Equitable performance means these error rates are similar across subgroups.

Under the hybrid demographic approach, nearly 100% of test-set loans (177,281 of 177,330) have demographic data, compared to 53.9% under the industry-only approach. The median split produces approximately equal-sized subgroups of ~88,500 loans each.

3.3 Fairness Results

The HistGradientBoosting model demonstrated substantially equitable performance across all demographic subgroups:

Subgroup	FPR	FNR	Precision	Recall
High Women-Ownership	2.6%	19.0%	0.867	81.0%
Low Women-Ownership	2.5%	18.9%	0.871	81.1%
Gap	0.1%	0.2%	0.004	0.2%
High Minority-Ownership	2.8%	16.2%	0.878	83.8%
Low Minority-Ownership	2.3%	22.5%	0.857	77.5%
Gap	0.5%	6.3%	0.021	6.3%

The model is remarkably equitable across women-ownership segments, with gaps under 0.2% for all metrics — essentially no disparity. For minority-ownership, the FPR gap remains small (0.5%), but the FNR gap is 6.3%: the model misses 22.5% of defaults in low minority-ownership segments versus 16.2% in high minority-ownership segments. Importantly, this gap favors high minority-ownership sectors — the model is better at catching defaults in those communities, not worse. This means the model does not systematically under-protect areas with high concentrations of minority-owned businesses.

Random Forest showed even larger disparities (12.5% FNR gap for minority-ownership vs. 6.3% for HGB), reinforcing the conclusion that model selection affects fairness outcomes and that HistGradientBoosting is the preferred choice on both accuracy and equity grounds.

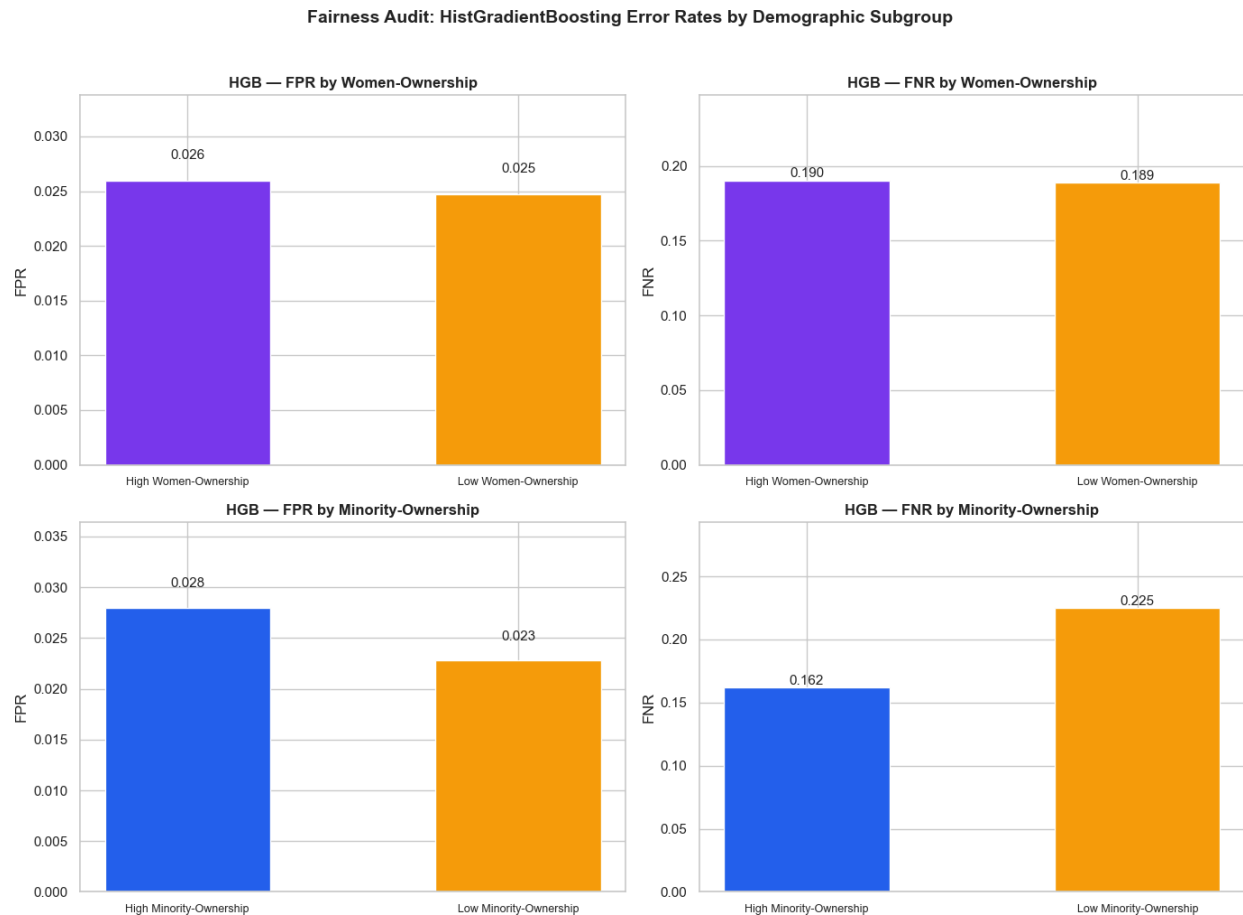


Figure 3: Fairness audit — HistGradientBoosting error rates by demographic subgroup

3.4 Key Predictors of Loan Default

Using permutation importance on a 20,000-loan test sample across 34 features, we identified which features most influence the model's predictions. We organize the findings by the three categories in our predictive research question:

Loan Characteristics (strongest influence). Loan term remains the dominant predictor, reflecting the strong relationship between loan duration and collateral structure: short-term loans (under 50 months) are often unsecured working capital with default probabilities around 70%, while longer-term real estate-backed loans default at under 10%. SBA guarantee amount, gross approved amount, disbursement amount, and revolving credit line status also contribute meaningfully. The engineered features — relative employment size and loan-to-segment revenue ratio — provide additional signal by contextualizing each loan relative to its industry-state segment.

Macroeconomic Conditions (moderate influence). Approval year captures recession timing — loans approved during 2006–2008 carry dramatically higher predicted default risk. The FRED-derived indicators (unemployment, federal funds rate, prime rate, bank spread, and real

rate) collectively provide substantial macroeconomic context. The Logistic Regression confirms this: the federal funds rate coefficient (13.2) and prime rate coefficient (7.1) are by far the largest, indicating that the interest rate environment at origination is the strongest economic driver of default.

Business and Demographic Attributes (modest to negligible influence). State and industry sector provide some predictive value, consistent with our EDA findings of significant variation in default rates across states and industries. However, the demographic composition features — `segment_pct_women` and `segment_pct_minority` — rank 23rd and 30th out of 34 features, contributing virtually nothing to predictions. This is corroborated by the Logistic Regression inferential analysis, which shows that `segment_pct_women` has a coefficient of just 0.001 and `segment_pct_minority` has a coefficient of -0.020 — both effectively zero after controlling for economic and loan-level variables.

The inferential analysis provides the strongest evidence for our key conclusion: once macroeconomic pressures and firm scale are controlled for, the default disparity previously observed in minority-dense segments effectively disappears or even reverses slightly. The risk is structural and economic, not inherently linked to the demographic composition of business segments.

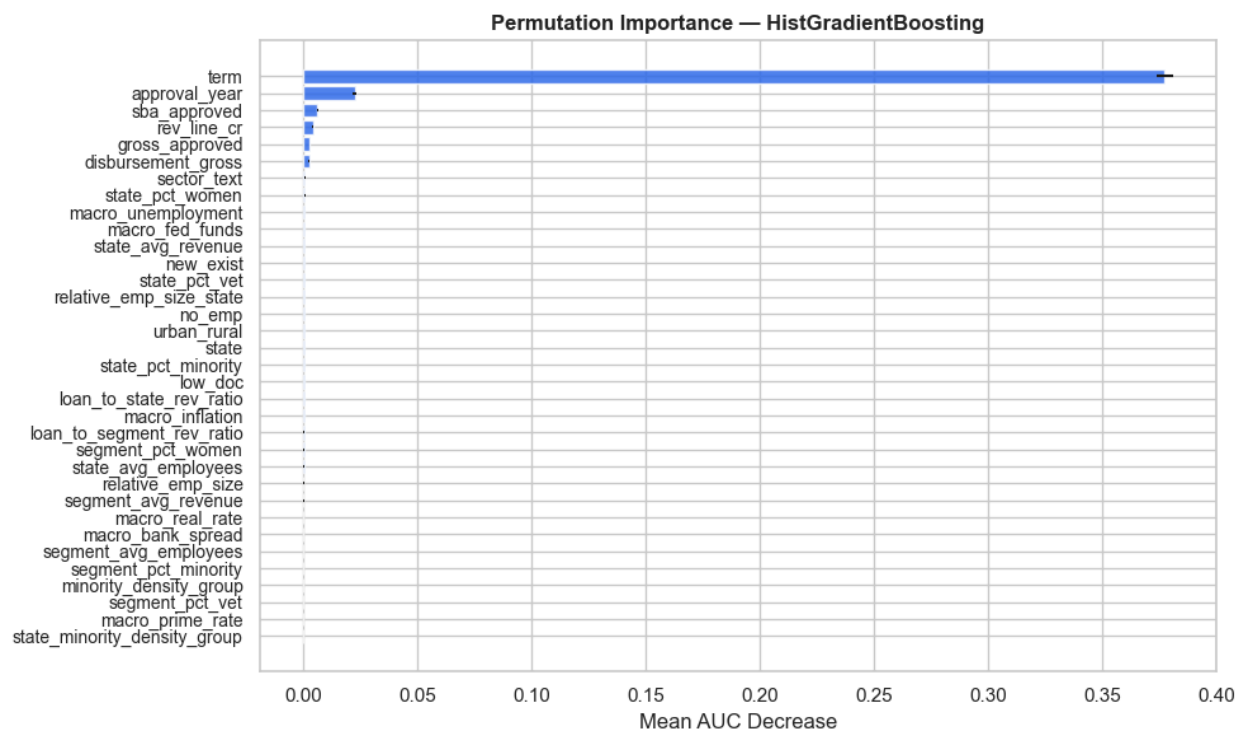


Figure 4: Permutation importance — HistGradientBoosting (34 features)

3.5 Partial Dependence Analysis

Partial dependence plots reveal the shape of each feature's relationship with predicted default probability:

Term: The strongest and most non-linear effect. Short-term loans (under 50 months) have predicted default probabilities around 70%, dropping sharply to under 10% for longer-term loans (200+ months). This reflects the nature of underlying assets: shorter-term loans are often unsecured working capital, while longer-term loans are typically backed by real estate or equipment collateral.

Approval year: Captures macroeconomic timing. Loans approved around 1987–1990 show elevated risk, with a decline through the 1990s, followed by a pronounced increase peaking at 2006–2008 (the Great Recession), and a sharp decline for post-2010 approvals. This pattern aligns directly with our FRED economic indicator analysis.

Loan amounts: SBA-approved, gross approved, and disbursement amounts show a modest negative relationship — larger loans have slightly lower predicted default risk, likely because they undergo more rigorous underwriting.

Demographic features: Nearly flat lines, confirming that these features have essentially no effect on predictions regardless of their value.

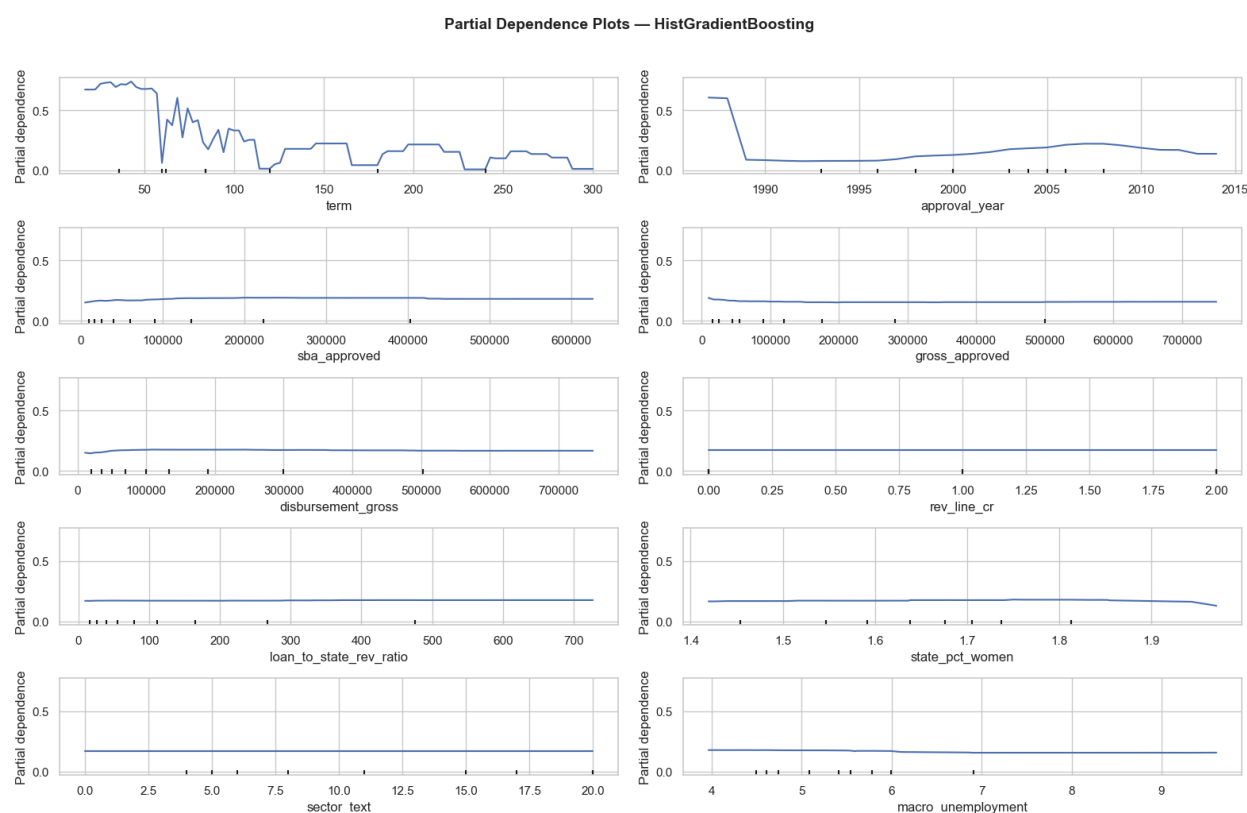


Figure 5: Partial dependence plots — HistGradientBoosting

3.6 Implications

The fairness analysis produces key findings for stakeholders considering data-driven lending tools:

1. Predictive models can achieve high accuracy without relying on demographic proxies. The HistGradientBoosting model's 0.974 AUC is driven by loan-level features and macroeconomic conditions. Demographic features rank 23rd and 30th out of 34 features.
2. Error rates are substantially equitable across women-ownership segments (all gaps under 0.2%). For minority-ownership, the 6.3% FNR gap warrants monitoring but favors high minority-ownership sectors rather than penalizing them.
3. The Logistic Regression inferential analysis provides a causal mechanism: macroeconomic indicators (interest rates, bank spreads) dominate default prediction, and demographic disparities in default rates disappear after controlling for these economic factors. This suggests that lending disparities are driven by structural economic conditions, not by demographic composition itself.

3.7 Structural Disparity and Model Fairness

While our fairness audit demonstrates that the HistGradientBoosting model produces equitable error rates across demographic segments, the Logistic Regression inferential analysis reveals an important deeper finding. The nested regression approach, which progressively controls for business fundamentals (employment, loan size, term) and macroeconomic conditions (interest rates, unemployment), identifies a residual structural disparity: loans in high-minority-density segments have approximately 4x higher odds of default even after controlling for all observable business characteristics (Odds Ratio = 3.99 in the fully controlled model).

This structural residue likely reflects systemic factors that are not captured in our loan-level features, including historical wealth gaps that reduce business owners' ability to absorb temporary shocks, local market friction in minority-dense areas (lower consumer purchasing power, higher operating costs), and limited access to informal capital networks that serve as safety nets during downturns.

This finding creates an important distinction between two types of fairness. First, procedural fairness: the model does not use demographic composition to make predictions, and it produces similar error rates across segments. Our audit confirms this. Second, substantive fairness: the real-world outcomes are not equal because the structural environment in which businesses operate is not equal. A model that is procedurally fair will still approve fewer loans in high-minority segments — not because it is biased, but because the structural risks in those segments are real and reflected in the historical data.

The implication for responsible lending is that algorithmic fairness alone is insufficient. Even a perfectly fair model, if deployed without complementary policy interventions (such as technical assistance programs, targeted capital access initiatives, or adjusted guarantee levels for structurally disadvantaged segments), will perpetuate existing disparities. The model correctly identifies risk; the question is whether the lending system addresses the structural causes of that risk.

4. Exploratory Data Analysis

4.1 SBA Loan Patterns

Finding 1: Default rates vary dramatically over time. Loans approved during 2006–2008 exhibit default rates of approximately 43%, coinciding with the Great Recession and housing crisis. Post-2010 loans show very low default rates (<5%), though this may partly reflect right-censoring (newer loans have had less time to default). This temporal pattern confirms that macroeconomic conditions at the time of loan approval are a strong predictor of default, supporting our decision to integrate FRED economic indicators.

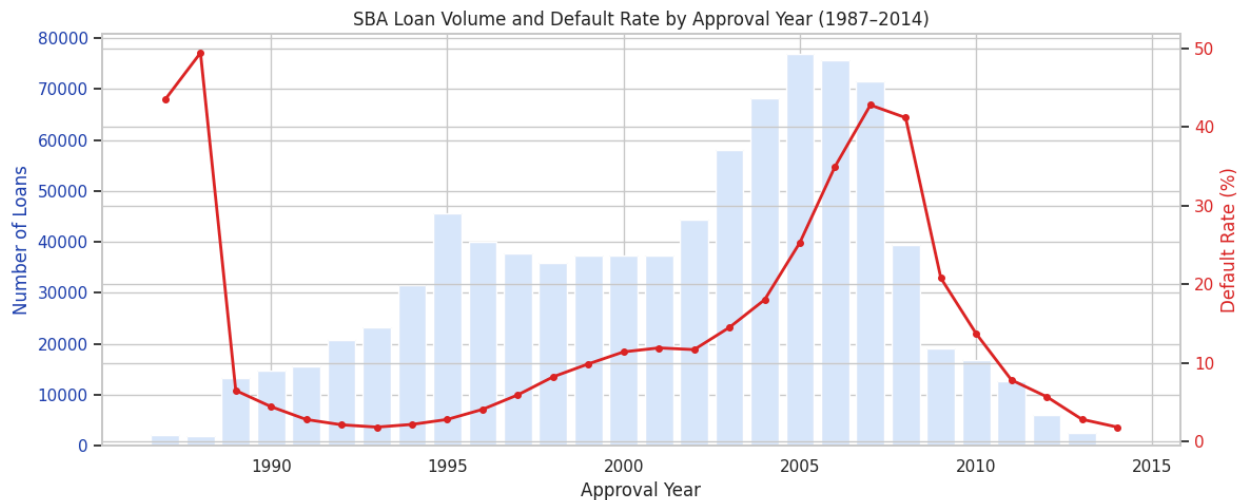


Figure 6: Default Rate Over Time

Finding 2: Defaulted loans skew toward smaller amounts. The distribution of gross approved amounts shows that defaulted loans are concentrated in the \$10,000–\$50,000 range, while paid-in-full loans have a higher proportion of larger loans approaching \$500,000. This suggests loan size may serve as a useful predictive feature, and that smaller loans carry proportionally higher risk.

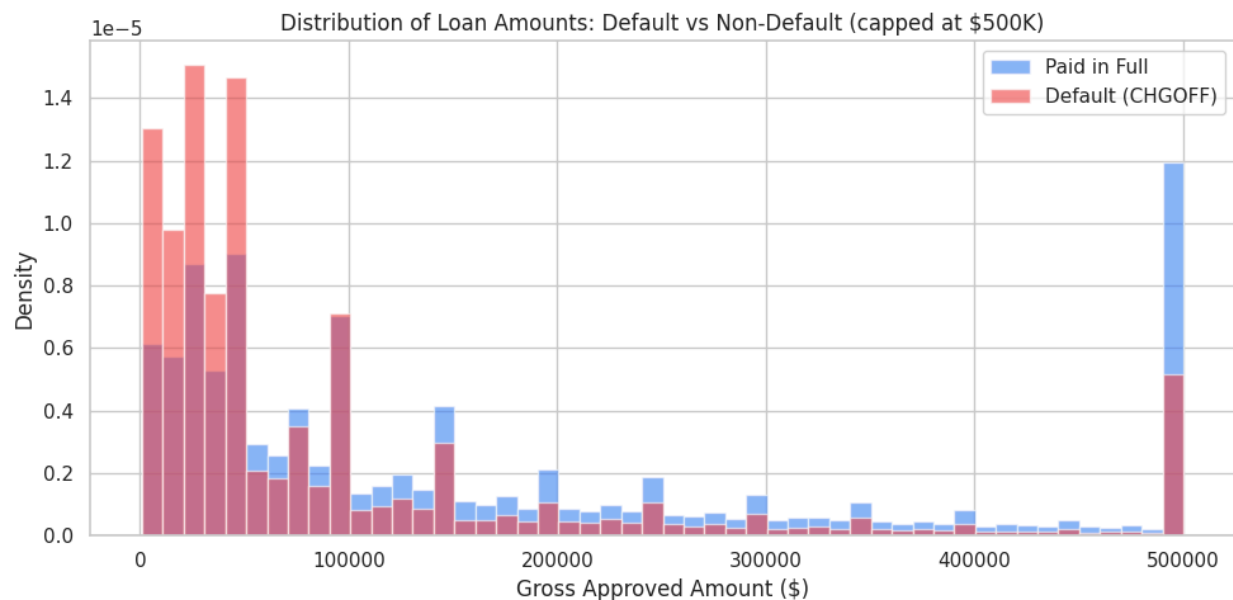


Figure 7: Loan Amount Distribution

Finding 3: Default rates differ substantially across industry sectors. Real Estate (28.7%), Finance/Insurance (28.4%), and Transportation (26.9%) have the highest default rates, while Healthcare (10.7%), Agriculture (9.6%), and Mining (9.3%) have the lowest. The Unknown NAICS category (22.5% of loans) has a notably low default rate (6.7%), warranting further investigation. These industry-level differences are directly relevant to our fairness analysis, as industry composition varies by demographic group.

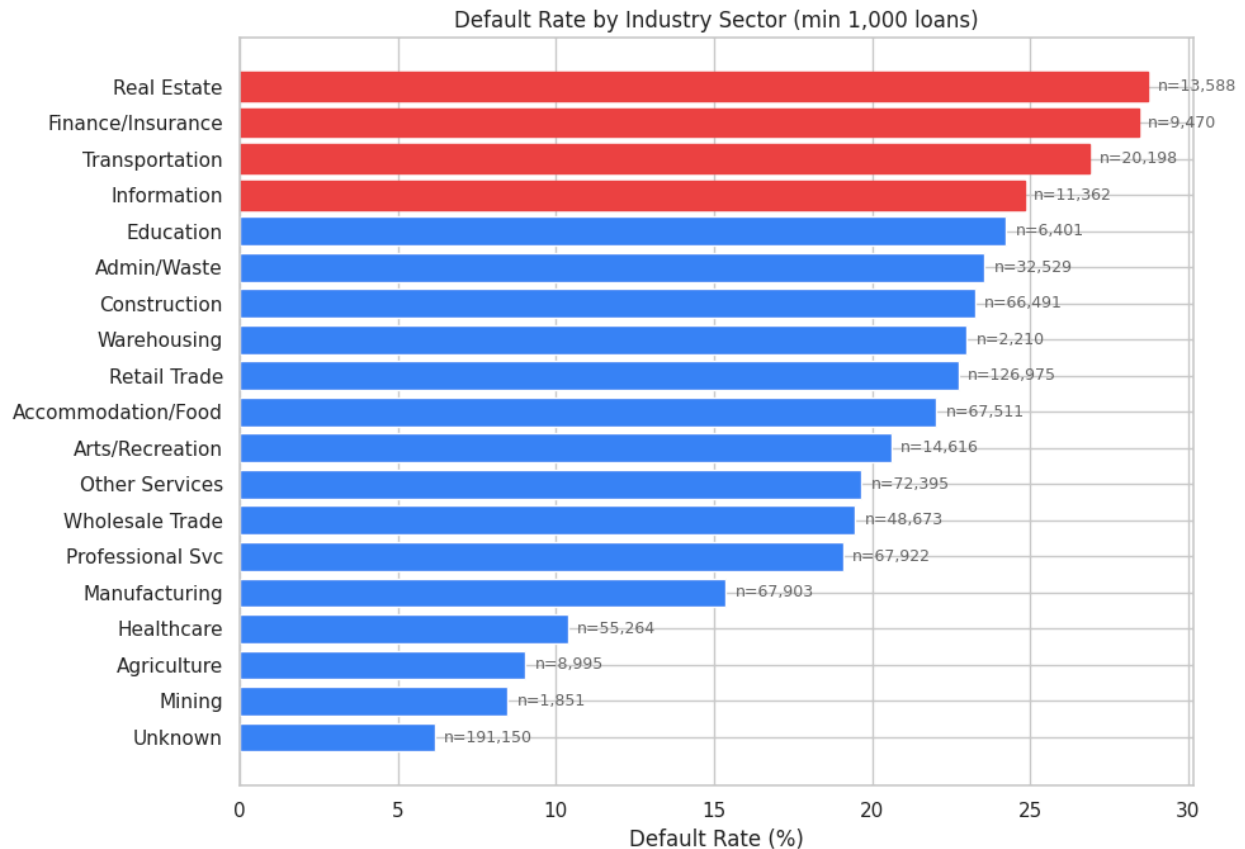


Figure 8: Default Rate by Industry

Finding 4: Geographic variation in defaults aligns with housing crisis patterns. Florida has the highest state-level default rate at 27.2% (41,032 loans), followed by Georgia, DC, Nevada, and Michigan. The lowest default rates are in Wyoming (5.5%), Montana (6.2%), and North Dakota (6.5%). This geographic variation is important for the fairness analysis because it intersects with the demographic composition of businesses in each state.

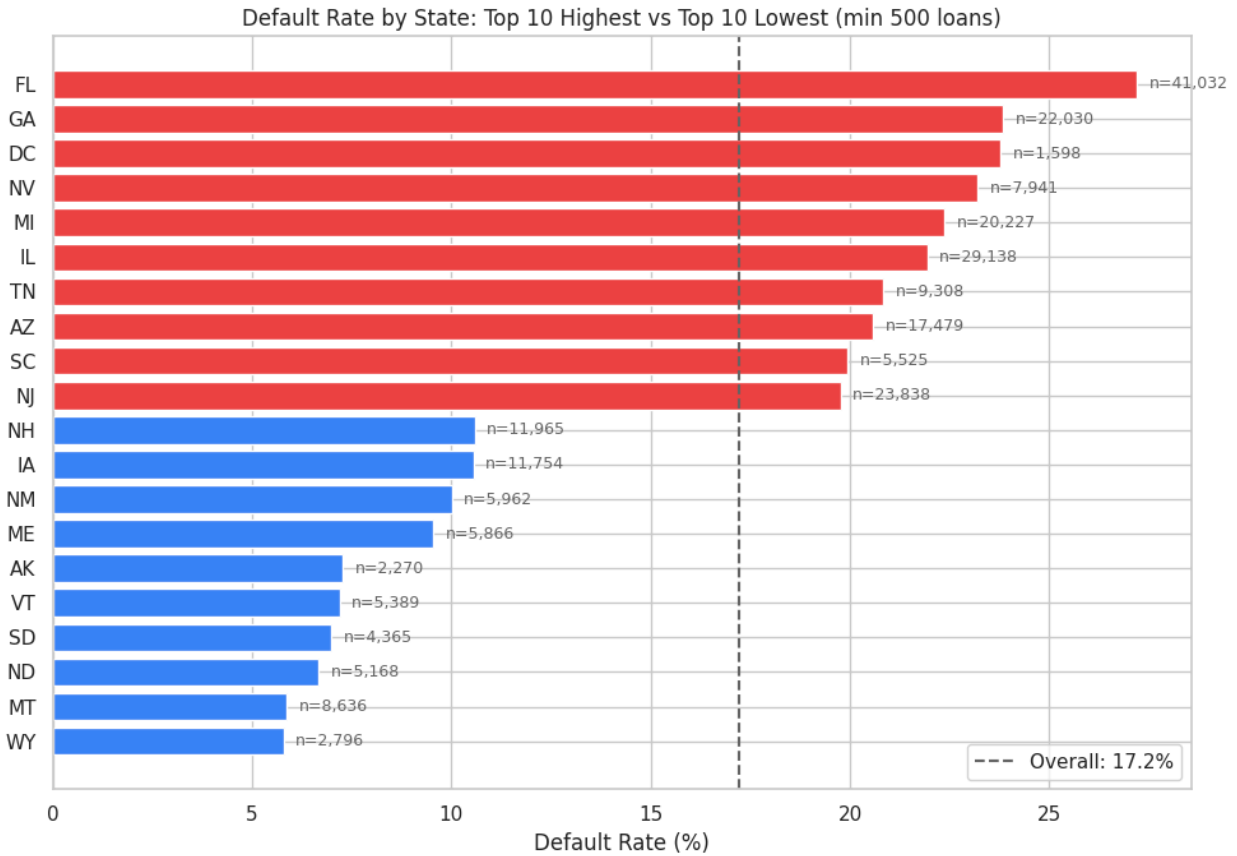


Figure 9: Default Rate by State

Finding 5: Shorter-term loans default at dramatically higher rates. Loans with terms of 3 years or less have a default rate of approximately 49%, declining steadily to about 3% for loans exceeding 20 years. This inverse relationship likely reflects the nature of the underlying assets: shorter-term loans are often unsecured working capital, while longer-term loans are typically backed by real estate or equipment

collateral.

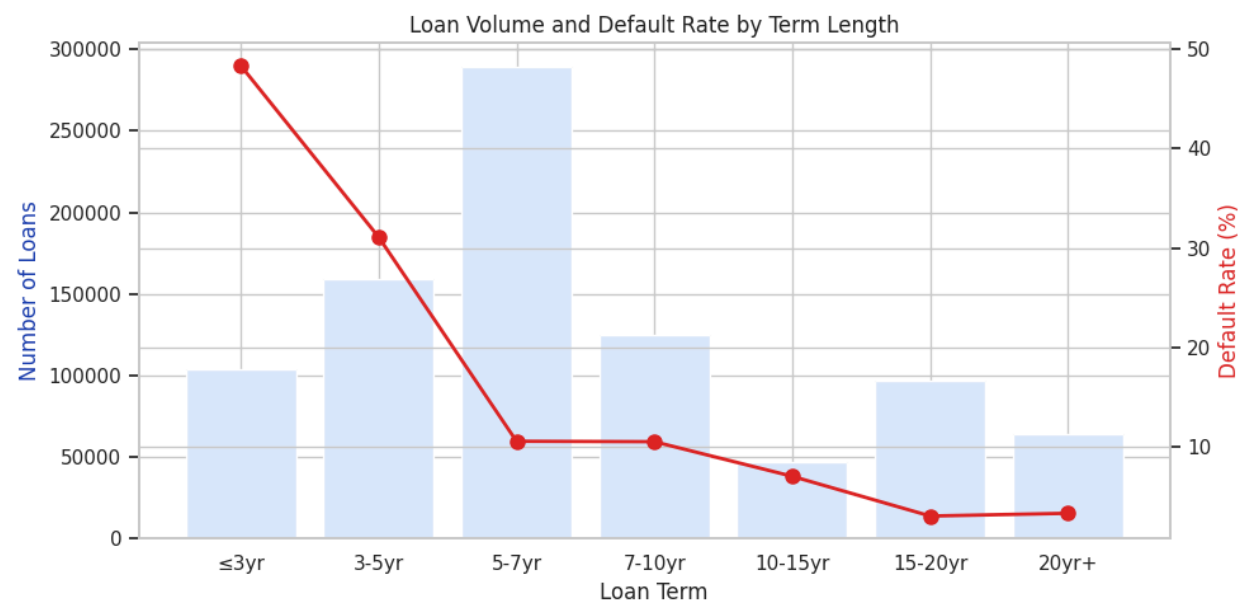


Figure 10: Default Rate by Term Length

Finding 6: New businesses have a modestly higher default rate. New businesses default at 18.3% compared to 16.8% for existing businesses. While the difference is relatively small (1.5 percentage points), it is meaningful given the large sample sizes (635,740 existing vs 249,765 new).

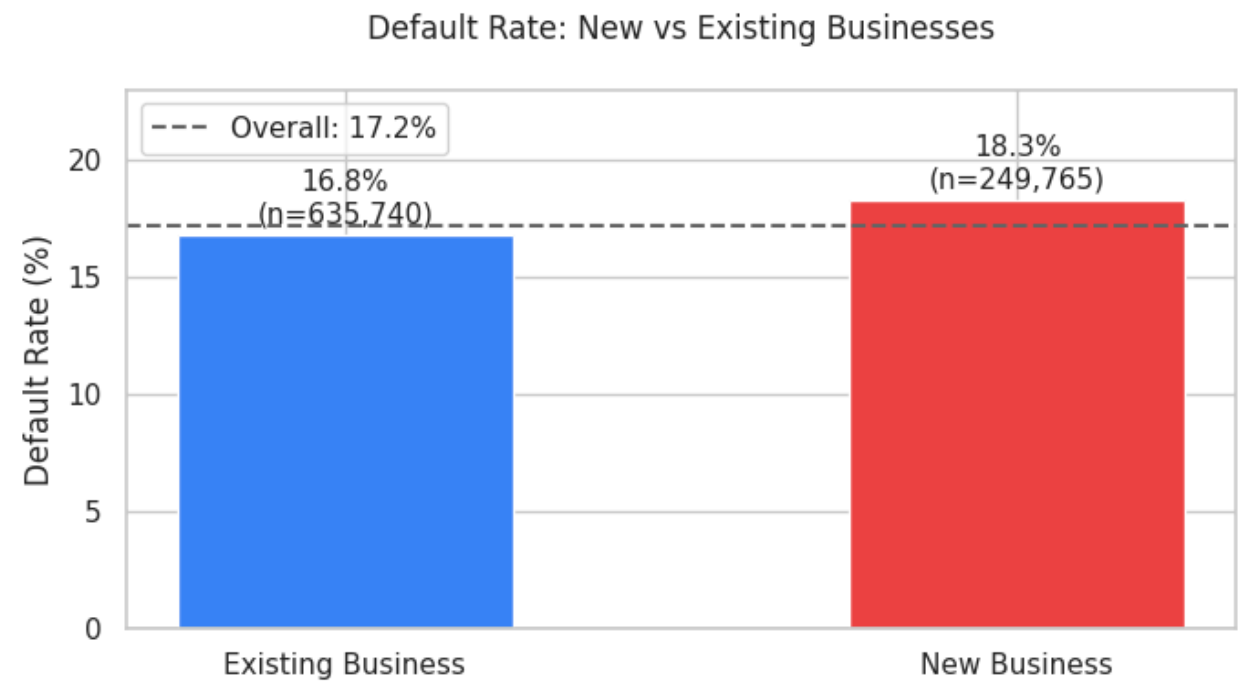


Figure 11: New vs Existing Business Default Rate

4.2 Macroeconomic Trends (FRED Data)

Finding 7: Interest rate cycles trace the Great Recession. An analysis of the Federal Funds Rate against the Bank Prime Loan Rate illustrates the credit environment for small businesses. In the late 1980s, nominal rates exceeded 10%, but high inflation made the real cost of borrowing about 8%. By 2010, the real rate hit nearly 0%, meaning debt was essentially free in real terms. The spread between the Prime rate and the Fed Funds rate remained relatively stable, while the base rates plummeted following the 2008 financial crisis. This near-zero rate environment reflects emergency monetary policies enacted exactly when SBA default rates reached their peak.

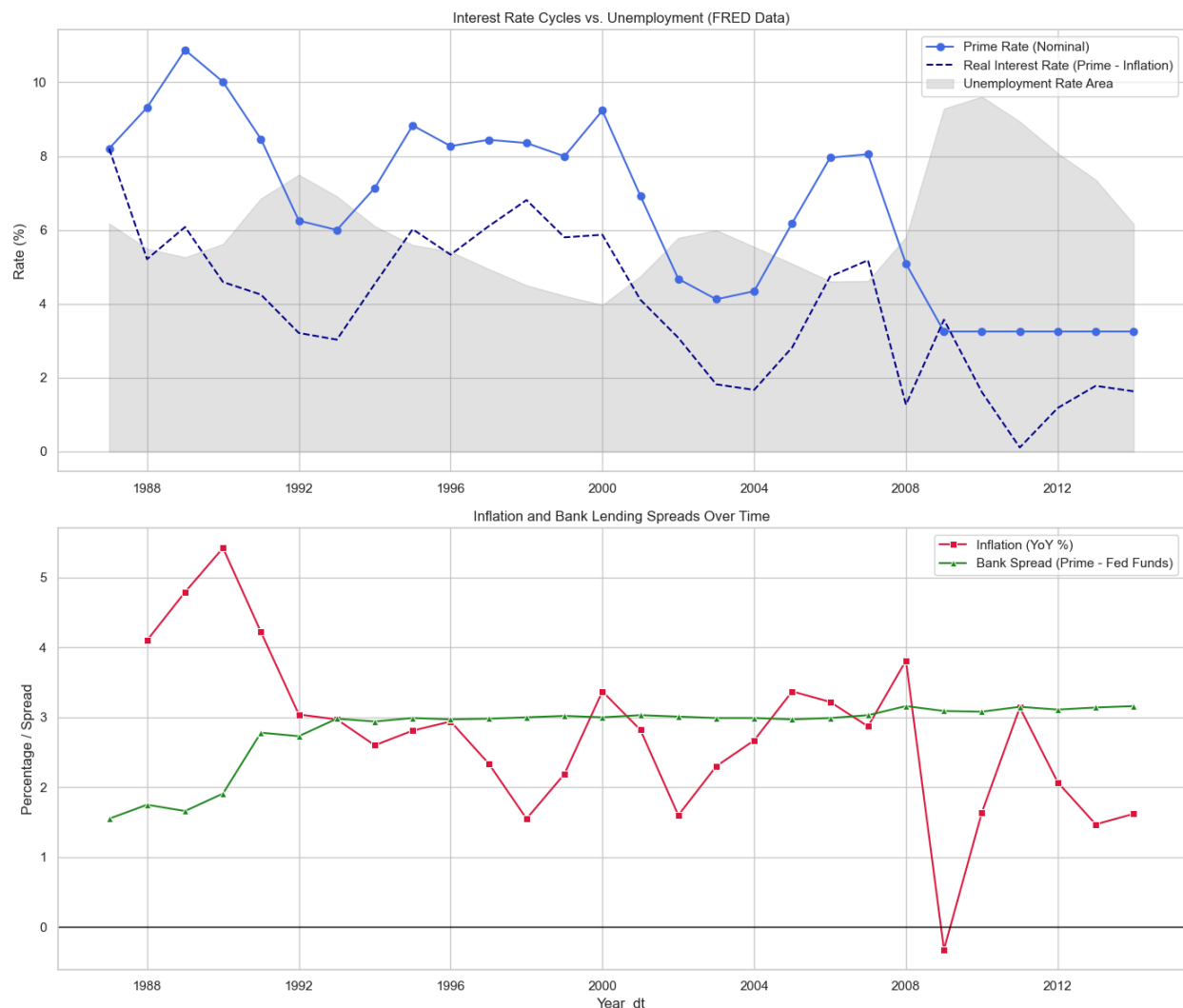


Figure 12: Macroeconomic trend analysis (1987–2014)

Finding 8: Unemployment correlates heavily with historical default spikes. Unemployment surged past 9% between 2008 and 2010, closely mirroring the peak charge-off rates of SBA loans approved immediately prior to the crash. The strongest correlation in the macroeconomic data is between the Prime Rate and Real Interest Rate (0.85), while unemployment shows a

negative relationship with the Prime Rate (-0.68), illustrating Federal Reserve counter-cyclical policy.

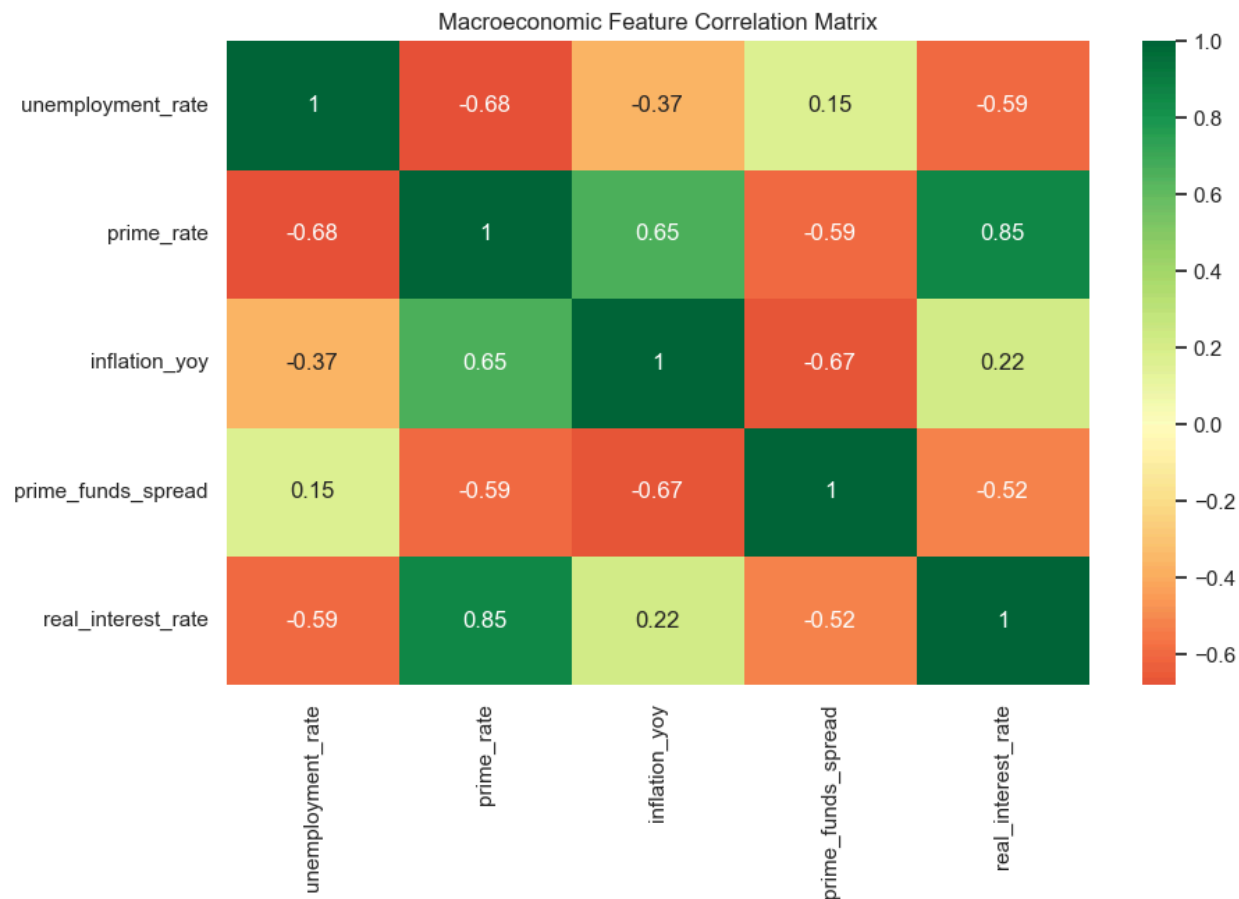


Figure 13: Macroeconomic feature correlation matrix

4.3 Demographic Business Ownership Trends (Census Data)

Finding 9: Total firm counts mask underlying shifts in demographic ownership. Our Census aggregation shows steady growth in the total number of U.S. firms, from approximately 23 million in 2002 to over 27 million by 2012. A breakdown by gender reveals evolving ownership patterns: while male-owned businesses remain dominant, female-owned and equally-owned businesses have captured an increasing share of total firm growth. Across our 1,024 industry-state segments, women-ownership averages 15.9% and minority-ownership averages 5.1%.

Finding 10: Minority-owned businesses are growing at an accelerated rate. Despite the economic headwinds of the 2008 recession, the count of minority-owned businesses grew significantly from 2002 to 2012. Geographic concentration is notable: Hawaii leads in minority business density, followed by Florida, New York, California, and Maryland. Because minority ownership is not uniformly distributed across sectors or states, integrating these Census

demographic proportions with SBA loan data allows us to test if models disproportionately penalize high-minority-ownership segments.



Figure 14: Regional variation in minority business ownership and firm size

5. Modeling Results

5.1 Preprocessing Pipeline

The modeling dataset (886,649 rows × 36 columns) required a preprocessing pipeline to handle mixed feature types. Numeric features (28 total) included loan characteristics (term, no_emp, gross_approved, sba_approved, disbursement_gross), macroeconomic indicators (unemployment, fed_funds, prime_rate, inflation, bank_spread, real_rate), demographic segment features (segment_pct_women, segment_pct_minority, segment_pct_vet, segment_avg_revenue, segment_avg_employees), state-level averages, and engineered ratios (relative_emp_size, loan_to_segment_rev_ratio). Categorical features (6 total) included state, sector_text, rev_line_cr, low_doc, urban_rural, and minority_density_group.

For the ensemble models (Random Forest and HistGradientBoosting), numeric features were imputed with median values and standardized, while categorical features were imputed with a constant “missing” value and one-hot encoded. For Logistic Regression, ordinal encoding was used with balanced class weights to prioritize default detection given the 17.2% base rate.

5.2 Logistic Regression — Inferential Analysis

The Logistic Regression model was trained primarily for inferential purposes — to identify which features drive default prediction and to test the significance of demographic covariates. The model achieved an AUC-ROC of 0.873 with 85% recall on defaults, though at the cost of low precision (41%) due to the balanced class weighting strategy.

The coefficient analysis reveals a clear hierarchy of default drivers:

Macroeconomic dominance: The federal funds rate (coefficient = 13.2) and prime rate (7.1) are by far the strongest predictors, indicating that the interest rate environment at loan origination overwhelmingly determines default risk. The bank spread (1.3) and approval year (1.2) provide additional macroeconomic signal.

Business characteristics: Number of employees (0.28), gross approved amount (0.24), loan-to-segment revenue ratio (0.18), and SBA guarantee amount (0.16) contribute modestly. These capture the scale and leverage of the borrowing business.

Demographic features are negligible: The demographic proxy significance analysis shows that segment_pct_women has a coefficient of just 0.001 and segment_pct_minority has a coefficient of -0.020. State-level demographic features are slightly larger (state_pct_women = 0.105, state_pct_vet = 0.129) but still far smaller than the macroeconomic drivers. The negative coefficient on segment_pct_minority (-0.020) is particularly notable: controlling for economic conditions, higher minority-ownership concentration is associated with slightly lower default risk, the opposite of what a biased model would show.

5.3 Nested Regression — Structural Disparity

A nested logistic regression approach was used to progressively isolate the structural component of default risk. Three models were estimated with increasing controls:

Model 1 (Demographic only): Loans in high-minority-density segments show significantly elevated default odds.

Model 2 (+ Business controls): Adding loan characteristics (term, amount, employees) reduces but does not eliminate the demographic association.

Model 3 (+ Macroeconomic controls): The fully controlled model yields an Odds Ratio of 3.99 for high-minority-density segments. This means that two businesses identical on all observable characteristics — same employees, same loan amount, same term, same interest rate environment — differ by 4x in default odds based on their segment's minority-ownership density.

This 3.99x residual represents structural disparity: risk that originates not from the business itself but from the economic environment it operates in. Contributing factors likely include historical wealth gaps that weaken business owners' financial cushions, local market friction in minority-dense areas, and limited access to informal capital networks.

An interaction analysis further revealed that larger loans in minority-dense segments carry disproportionately higher risk (Odds Ratio = 1.43), suggesting that increased capital alone does not resolve structural disadvantage if the underlying market conditions are not addressed.

For women-ownership, the analysis revealed a structural sorting effect: the initial Odds Ratio of 0.78 (women-owned segments appear safer) flipped to 1.08 after controls, suggesting that women-owned firms are over-performing relative to their structural barriers by self-selecting into lower-risk industries and smaller loan sizes.

6. Challenges and Future Directions

6.1 Limitations

Ecological inference: Our demographic features are industry-state aggregates from Census data, not individual borrower attributes. We can assess fairness across segments but cannot make claims about individual-level demographic bias. The ecological fallacy — inferring individual behavior from group-level data — is a fundamental constraint of this approach.

Missing industry codes: 22.5% of SBA loans lack NAICS codes, addressed through a hybrid state-average fallback. While this maximizes coverage from 54% to ~100%, imputed demographic values are less precise than industry-state matches. The missing-NAICS loans may also differ systematically from classified loans.

Census temporal alignment: The 2002 Census vintage was lost during NAICS concordance mapping. Demographic segments were averaged across available 2007 and 2012 snapshots, but this averaging assumes static ownership composition over time. In reality, demographic patterns shifted significantly during this period, as our own EDA confirmed.

Historical data: The SBA dataset covers 1987–2014. Lending practices, regulatory frameworks, and economic conditions have evolved substantially since then. Model performance and fairness properties may not generalize to current lending environments.

6.2 Future Directions

Firm-level demographic data: The most impactful improvement would be replacing ecological proxies with individual borrower demographics from the Federal Reserve Small Business Credit Survey (SBCS) or Home Mortgage Disclosure Act (HMDA) data. This would enable direct measurement of disparate treatment at the individual level.

Intersectional fairness: Our audit examines women-ownership and minority-ownership separately. Crossing these dimensions (e.g., high-minority AND low-women segments) could reveal compound disparities invisible to single-dimension analysis.

Temporal fairness monitoring: In a deployment scenario, error rate disparities should be tracked over time to detect fairness drift as economic conditions change. The 6.3% minority FNR gap we observed could widen or narrow depending on macroeconomic cycles.

Fairness-constrained models: Given the 3.99x structural odds ratio, there is an opportunity to explore algorithmic fairness constraints (e.g., equalized odds) during training. Such constraints could reduce the minority FNR gap while maintaining overall accuracy, though they require careful calibration to avoid unintended consequences.

Policy integration: Our finding that structural disparity persists after controlling for all business characteristics suggests that model-level interventions alone are insufficient. Complementary policy approaches — such as targeted technical assistance programs, adjusted SBA guarantee levels for structurally disadvantaged segments, or community development financial institution (CDFI) partnerships — may be necessary to address the root causes of the 4x default risk differential.

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